SOFTWARE ENGINEERING LABORATORY

LAB–2

USE CASE DIAGRAM AND USE CASE SPECIFICATIONS

LIBRARY MANAGEMENT SYSTEM

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Use Case Modelling – Library Management System

1. Aim

To identify the actors and use cases of a Library Management System, draw its use case diagram, and specify the role, function, precondition, and postcondition of each use case.

2. Introduction

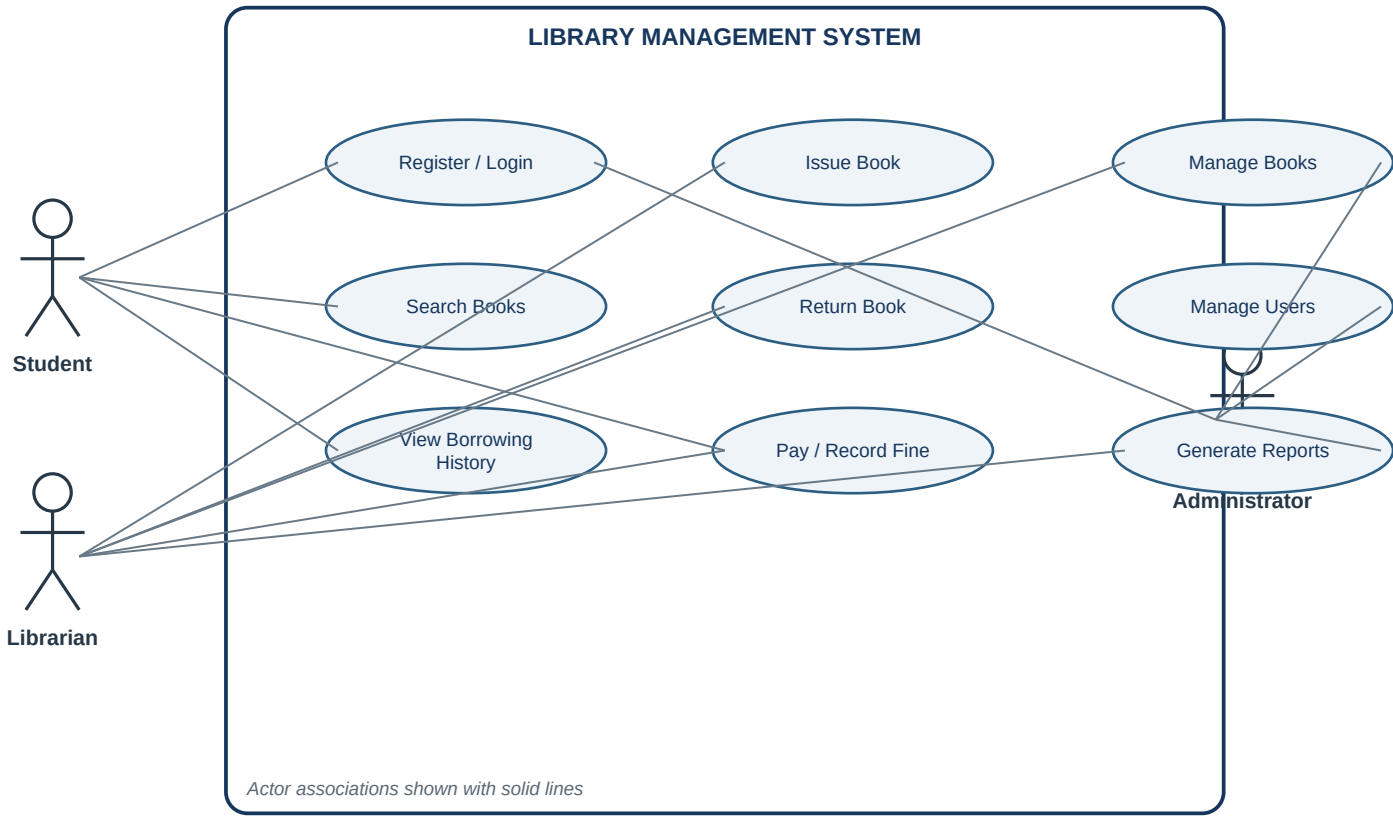
A use case diagram is a UML (Unified Modeling Language) behavioural diagram that represents how external users (actors) interact with a system. It describes the system's major services from the user's point of view. In a Library Management System, the main actors are Student, Librarian, and Administrator.

3. Actors and Their Roles

Actor	Role / Responsibilities
Student	Registers or logs in, searches the book catalogue, checks availability, views current and previous borrowing records, and views or pays pending fines.
Librarian	Maintains book records, issues books to students, records returned books, handles fine payments, and generates circulation-related reports.
Administrator	Manages user and librarian accounts, maintains system-level settings, monitors system activity, and accesses overall reports.

4. Use Case Diagram

The following diagram shows the major interactions between the actors and the Library Management System. Solid lines represent associations between actors and use cases.



5. Use Case Specifications

Each use case below describes its function, precondition, and postcondition. Preconditions are the conditions that must be true before the use case begins. Postconditions describe the state of the system after successful completion.

UC-01: Register / Login

Specification	Description
Function	Allows a student, librarian, or administrator to create an account where applicable and securely access the system.
Precondition	For registration, the user must not already have an account with the same unique ID or email. For login, the user must have an active registered account.
Postcondition	A valid account is created or the user is authenticated and redirected to the dashboard appropriate to their role. Invalid credentials are rejected.

UC-02: Search Books

Specification	Description
Function	Allows students and librarians to find books by title, author, ISBN, or category and view availability.
Precondition	The Library Management System is running. Login is required if catalogue access is restricted by the library.
Postcondition	Matching books and their details are displayed. The system does not modify book or transaction records.

UC-03: View Borrowing History

Specification	Description
Function	Allows a student to view currently issued books, due dates, previously returned books, and related transaction details.
Precondition	The student is logged in and has a valid library membership.
Postcondition	The student's borrowing history and current loan status are displayed. No transaction data is changed.

UC-04: Issue Book

Specification	Description
Function	Allows a librarian to issue an available book to an eligible registered student and record the issue and due dates.
Precondition	The librarian is authenticated; the student has a valid membership; the book exists and has at least one available copy; the student has not exceeded the borrowing limit.
Postcondition	A new issue record is created, the due date is stored, and the available quantity is reduced by one. The student can see the issued book in their account.

UC-05: Return Book

Specification	Description
Function	Allows a librarian to record the return of a previously issued book and close the loan transaction.
Precondition	The librarian is authenticated and the book has an active issue record for the student.
Postcondition	The return date and transaction status are updated, and the available quantity is increased by one. If overdue, the applicable fine is calculated or recorded.

UC-06: Pay / Record Fine

Specification	Description
Function	Allows a student to pay a pending fine through an enabled payment method, or a librarian to record payment received at the library.
Precondition	A valid fine record exists and is unpaid. The student or librarian is authenticated with the required permission.
Postcondition	The payment is recorded, the fine status is updated to Paid, and a payment reference or receipt is made available where supported.

UC-07: Manage Books

Specification	Description
Function	Allows a librarian to add new book records, update book information or quantities, and remove records when permitted.
Precondition	The librarian or administrator is authenticated and has book-management permission. The book details supplied for an update or addition are valid.
Postcondition	The book catalogue is updated. Changes are saved and can be seen in subsequent searches. Existing transaction history is preserved.

UC-08: Manage Users

Specification	Description
Function	Allows an administrator to create, update, deactivate, or manage student and librarian accounts and assign appropriate roles.
Precondition	The administrator is authenticated and has user-management permission.
Postcondition	The user account information and role assignments are saved. Deactivated users can no longer access protected system functions.

UC-09: Generate Reports

Specification	Description
Function	Allows librarians and administrators to generate reports such as issued books, returned books, overdue books, inventory, and registered users.
Precondition	The librarian or administrator is authenticated and has reporting permission. Relevant records exist in the system.
Postcondition	The requested report is generated and displayed or exported. Report generation does not modify the underlying records.

6. Conclusion

The use case diagram identifies the main actors and the services they use in the Library Management System. The Student interacts mainly with book searching, borrowing history, and fine-related services. The Librarian manages books and circulation activities, while the Administrator manages users and system-level operations. The use case specifications define the function, preconditions, and postconditions of each major operation and provide a clear basis for system design and testing.

END OF LAB–2